◊Unknown

Thoughts at The End of Time

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1. Introduction

It is always logically possible that something exists which no agent knows. A salient law of possibility has shown itself prevalent throughout logic. A law that is infinitely rooted in the nature of knowledge. Across logic, epistemology, and metaphysics, a law has remained unexamined and largely overlooked. The law of ◊Unknown (Possibility Unknown). ◊Unknown expresses the laws in which the space of possible unknowns is logically infinite, structurally open, and incapable of final closure. Rather than treating the unknown as a temporary absence of knowledge, ◊Unknown is a permanent feature of modal reality. ◊Unknown states the fundamental law: that the openness of the unknown is not just an extraneous null, but is a logical possibility and is necessary. ◊Unknown is a logically coherent system that widens logic's understanding of logical possibilities.

Why does ◊Unknown matter? At first glance, ◊Unknown may seem like objective logic's metaphorical paranoia, but it is much more than that. With any truth claim, objective logic expresses certainty by defining what is. Except that no one can logically claim that anything is absolutely true if the unknown exists. Our ignorance is evidence of ◊Unknown. Any and every truth statement is subject to the infinite possibilities of the unknown. This infinite continuity begs the question of "*What if?*" ◊Unknown greatly revolves around the question of "*What if?*", not in a sense of denial, but in a sense of epistemic openness. An example of this "*What if*" factor could be taking "*2+2=4*", and a response of inquiry from ◊Unknown could possibly look like, "*What if there is something that is outside your knowledge that disproves this claim?*" Although "*2+2=4*" is an objective truth to us and is logically sound, the possibility of it not being true due to unknown factors is also logical. The importance of ◊Unknown comes from its influence on how conclusions should be drawn. Instead of "2+2=4", ◊Unknown would expand this statement to, "*2+2=4 is the most probable objective truth from our knowledge*". Obviously, this is not how it should be said every time, but this example helps show how ◊Unknown builds a foundation for

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its epistemic openness in a sufficient introductory sketch. That even for objective truths, ◇Unknown has the possibility of logically testing them. On the other hand, the possibility of this outcome is not probable, which helps us differentiate between possible outcomes of ◇Unknown and probable outcomes of ◇Unknown.

So, how does ◇Unknown not drive us insane? It may seem at first that the logical possibility of infinite unknowns would undermine knowledge itself or lead to a kind of epistemic despair. If an infinitesimal amount of possible unknown truths exists beyond any system, why even have a system in the first place? However, this approach mistakes openness for futility. ◇Unknown does not state that knowledge is impossible, but that it is never final. In fact, the very incompleteness of the truth is what drives inquiry forward. This unknown is what makes science thrive, philosophy advance, and thoughts deepen. This motivation from the unknown is the recognition that there is always more to understand. ◇Unknown secures that the drive is not just a feeling or assumption, but is a logical necessity. ◇Unknown liberates the knower from the illusion of finality and replaces it with a framework of figurative "humility", where knowing is real, but always provisional, and progress is meaningful precisely because the frontier is endless. Just as a compass gains meaning only in a world with uncharted regions, so does logical thought gain meaning in a universe where possibility exceeds possession. ◇Unknown secures the law that rationality never collapses into arrogance or despair, but remains an open pursuit of what may yet be.

1.1 Possibilities and Probabilities within◇Unknown

◇Unknown also helps expand our understanding of the relationships between possibilities and probabilities, two concepts that are often conflated in the reasoning of a wide variety. Possibility is a modal concept: it concerns what can be, independent of how likely it is. In contrast, probability is a measure of how likely something is to occur within a defined framework of outcomes. ◇Unknown clarifies that possibility is logically prior to probability. Before probability can be assigned to a truth, we must assume its probabilistic scope. In this way, ◇Unknown helps to define the horizon of what may be considered in any probabilistic model. It conveys that not all possibilities are probabilistically quantifiable, especially when they lie beyond the current boundaries of what is known. By doing so, we prevent our understanding of

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the unknown from collapsing into closed statistical systems and open space for reasoning to expand. The infinite nature of ◊Unknown is not irrational or illogical, but is permissible and could even be inevitable. Just because a proposition is illogical does not mean that its possibility is illogical. An illogical possibility can be logically possible because the unknown permits it. It may not be probable, but it always has a factor of possibility. Furthermore, the extent of the possibility's illogical nature can help determine its distance from probability.

◊Unknown expresses possibility and is a modal claim. As such, ◊Unknown operates outside the boundaries of traditional probability theory, which relies on defined sample spaces, measurable distributions, and even frequency. Probability concerns its measurements of uncertainty within a known domain, while ◊Unknown spans across the logical openness of infinite unknown domains. Within probabilistic reasoning, all possible outcomes must be known in advance. Except that ◊Unknown makes this impossible to accomplish. This proposes that new, true propositions may always exist beyond any system's horizon, making the total event space unknowable in principle. Therefore, attempting to assign a statistical probability to the existence of unknowns would either beg the question or undermine the law of ◊Unknown itself. If we could fully enumerate the possibility space, then by definition, those truths would no longer be "unknown"; they would simply be unverified.

That being said, probability can still be assigned and used as an instrumental tool within the infinite domain of ◊Unknown; there cannot, however, be a finite statistic. Remember that ◊Unknown doesn't render our efforts to understand meaningless, but it eliminates final closure. So, probability can still be found within the constraints of our own knowledge, but these probabilities must be left open-ended. This does not mean that closed-ended probabilities do not exist, because we can still find probabilities within the finite domains of our knowledge, excluding ◊Unknown. This does not make statistical probabilities illogical or wrong, but just shows that objective logic can be used within the domain of our knowledge and not only the infinite span of ◊Unknown. For example, one could reasonably estimate the likelihood that unknown laws of physics remain undiscovered, or apply Bayesian models to gauge the probability of gaps in a machine learning system's knowledge. These probabilistic approaches model our degree of knowledge in the presence of unknowns. In this sense, statistics can help us

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illustrate the consequences of epistemic openness, but cannot fully capture its modal foundation. ◇Unknown resists quantification, not due to a lack of rigor, but because its truth lies in the logical necessity of its infinite openness, not in empirical uncertainty. Its non-quantifiability is not a weakness, but a reflection of the very principle it articulates: that there may always be more beyond what is measurable, quantifiable, conceivable, and or known; and this possibility cannot be collapsed into probability.

1.2 ◇Unknown from a Contextual Perspective

◇Unknown's scope of knowledge has deep roots in modern logic and philosophy, but has not been formalized in modal logic prior to now. In 1931, Kurt Gödel logically found that in any consistent formal system powerful enough to express arithmetic, there are true statements that are unprovable within that system. Gödel shattered the Hilbertian dream of mathematical completeness, demonstrating that truth outruns provability. This result leads directly to the core claim of ◇Unknown, that there is always a possibility that some truth lies outside the reach of any knower or system. Although Gödel's theorems apply specifically to formal mathematical systems, they still show elements of ◇Unknown throughout logic's history. And where Gödel shows incompleteness in logic, ◇Unknown shows open-endedness in knowledge itself, across all domains.

Karl Popper argued that science progresses not by verification, but by falsifiability, the idea that a claim is scientific only if it can, in principle, be proven false. Popper viewed knowledge as always provisional and subject to revision when confronted with new evidence. ◇Unknown expands upon this idea that new evidence can always logically emerge, even if we do not currently know what form it will take. Whereas Popper focused on empirical falsification, ◇Unknown introduces a modal layer: not only do we not know everything, but it is logically necessary that we never can. It shifts epistemic openness from a methodological caution to a law governing the structure of knowledge itself.

In addition, Immanuel Kant had also highlighted elements of ◇Unknown. ◇Unknown can be somewhat spotted with Immanuel Kant's study of the noumenal. Immanuel Kant distinguishes between the phenomenal world (what we can experience and know) and the noumenal world

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(things as they are in themselves, beyond human cognition). While Kant's framework was metaphysical, ◇Unknown reflects a similar idea: that there may always exist truths that transcend any given epistemic framework. Yet, ◇Unknown avoids metaphysical dualism by stating this not as a division between "what is" and "what is perceived", but as a modal feature of all epistemic structures. In this sense, ◇Unknown is somewhat Kantian in concept, but logically neutral in application; it applies to minds, machines, and systems alike.

Both Alan Turing's halting problem and Gregory Chaitin's algorithmic information theory demonstrate limits in computation: that some problems are undecidable, and some strings are algorithmically incomprehensible. These results are more than technical; they represent real barriers to complete knowledge, even purely formal systems. ◇Unknown translates these insights into a universal epistemic law: any system's view of truth is necessarily incomplete, not because of empirical gaps, but because of its logical structure. What Turing and Chaitin show in computation, ◇Unknown reveals across all knowledge-generating agents.

The previous references were not just sprinkled with elements of meaningless ignorance, but reflected how ◇Unknown works within logic without anyone taking full note of its stealthy self-application. Even without the name and identification of ◇Unknown, this law still makes itself present and known throughout the work of logic. ◇Unknown isn't just an idea or theory, but is a law that is necessary for comprehending and widening the scope of logical philosophy. While Gödel, Popper, Kant, and others helped expose the limits of certainty, their frameworks remained domain-specific or metaphysically bounded. ◇Unknown unifies these scattered insights into a single modal-epistemic principle: that the logical possibility of unknown truths is not a weakness of systems or subjects, but a law of knowledge itself. It does not replace these thinkers; it somewhat clarifies and extends their teachings into a logical framework that captures the full scope of epistemic incompleteness.

2. Formalization of the Unknown

Why does ◇Unknown work? As much as the previous explanations of ◇Unknown have been logically sound, to establish ◇Unknown not merely as a philosophical insight but as a formally entailed logical truth, a presentation of its structure within modal-epistemic logic is

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required. Prior to ◇Unknown's proofs, a slight clarification is to be made to a proposition we use to fulfill the horizons of ◇Unknown. When '$T(p)$' is used, it refers to a true proposition. Now, with this truth, it does NOT mean that the proposition is a fact, but defines that the proposition is true to be known; that an agent can epistemically take in and recall the information of the proposition to the degree of said agent's ability, not inclusive of temporal points. For further explanation, a proposition can be something of a false claim and still be truthfully known. The knowledge of the proposition is true, but the information of the proposition is not relevant to the truth of its knowledge within these proofs.

Formal Statement: $\forall x \Diamond \exists p \notin P \wedge K_x(T(p))$

x**:** any epistemic agent (human, AI, deity, etc.)

p**:** a proposition

P**:** the set of all truth-bearing propositions

K_x**:** the set of propositions known by agent x

$T(p)$**:** means "p is true to be known"

◇**:** modal operator for logical possibility

This statement reads, for any agent x, it is possible that there exists a true proposition p that x does not know.

Modal Table:

World	$T(p_1)$	$K_x(p_1)$	$T(p_2)$	$K_x(p_2)$	$T(p_3)$	$K_x(p_3)$	◇Unknown
w_1	T	T	T	F	F	F	Yes- p_2 is true, but unknown
w_2	T	T	T	T	T	F	Yes- p_3 is true but unknown
w_3	T	T	T	T	T	T	No- all truths are known

In w_1: p_2 is true and not known → ◇Unknown is satisfied

In w_2: p_3 is true and not known → ◇Unknown is satisfied

In w_3: All true propositions are known → ◇Unknown is not satisfied in this world alone

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EXCEPT, ◇Unknown only requires that it is possible, so even if you're in a world where everything is known, if another accessible world contains an unknown truth, then ◇Unknown still holds from the modal standpoint.

Secondary Entailments: $\Diamond[\neg K_x(p) \wedge T(p)] \rightarrow \neg\Box K_x(p)$

"If unknown truths are possible, then it cannot be necessarily true that an agent knows everything."

Deductive Proof Sketch: $K_x \subseteq P$

$\exists p \in P \wedge p \notin K_x$ is logically possible

Therefore: $\Diamond\exists p \notin K_x \wedge T(p)$

Therefore, ◇Unknown holds for all x

Modal Axiom (S5): $\Diamond T(p) \rightarrow \Box\Diamond T(p)$

So, ◇Unknown isn't just "maybe sometimes true"; it's modal stable across all logically consistent frames.

◇**Unknown's Infinite Nature:** $\forall x \Diamond \exists p(T(p) \wedge \neg K_x(p))$

$\forall x \forall \mathrm{n} \in N, \Diamond \exists P_n(|P_n| = n \wedge \forall p \in P_n \wedge T(p) \wedge \neg K_x(p))$

For any number n, it's possible that there are at least n unknown truths

$\forall x \Diamond \exists P_\infty(|P_\infty| = \infty \wedge \forall p \in P_\infty, T(p) \wedge \neg K_x(p))$

It is logically possible that an agent never exhausts the space of truths they do not and cannot know.

$[\forall x \Diamond \exists p\{T(p) \wedge \neg K_x(p)\}] \vDash [\forall x \Diamond \exists P_\infty\{|P_\infty| = \infty \wedge \forall p \in P_\infty \wedge T(p) \wedge \neg K_x(p)\}]$

3. Counterexamples

Falsifier: ◇Unknown creates a falsifier for itself because ◇Unknown makes the possibility of ◇Unknown being negated

$\neg\forall x\{\Diamond\exists p \notin P \wedge K_x[T(p)]\} \vDash \Box\forall p \in P \wedge \{T(p) \rightarrow K_x(p)\}$

"It is necessarily the case that every true proposition is known by some agent x."

This implies a self-falsifiability logic loop because ◇Unknown is fulfilling its own laws to falsify its own laws.

U**:** the proposition that ◇Unknown is false

K_x**:** the agent's knowledge

$T(U)$**:** "◇Unknown is false" is true

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$T(U) \wedge U \notin K_x$

Then, ◇Unknown is true, because there's a true proposition not known. So, if ◇Unknown is false and its falsehood is unknown, then ◇Unknown is true.

Formal Entailment Falsifier: $\neg \Diamond \exists p \notin K_x(T(p))$

U**:** "◇Unknown is false."

$T(U) \wedge U \notin K_x \rightarrow \exists p \notin K_x \wedge T(p)$

This contradicts the original assumption that ◇Unknown is false.

In the end, we derive: '$T(U) \wedge U \notin K_x \Rightarrow \neg U$' Any attempts to falsify ◇Unknown, if that falsifier is not universally known, become proof of ◇Unknown. This makes ◇Unknown a self-bolstering modal law: it logically entails the possibility of its own falsification, and this very possibility reaffirms the truth of this law. You need to fulfill ◇Unknown to even try to falsify it.

3.1 Counterarguments

Another falsifier would be an argument of "complete knowledge", suggesting omniscience, total determinism, idealism, and rationalism.

Formal Falsifier: $\exists x \Box \forall p: T(p) \rightarrow K_x(p)$

"There exists an agent x such that all truths are necessarily known by x." This is a claim that ◇Unknown directly challenges.

◇**Unknown's Logical Counter:** $\forall x \Diamond \exists p: T(p) \wedge \neg K_x(p)$

◇Unknown's laws do not allow for the necessity of complete knowledge. If someone asserts this argument, then they must have evidence that all knowns are finite to negate ◇Unknown's infinite nature. So, to accept complete knowledge, one must show that all truths are accessible, that all truths are already known (by at least one mind), and that no unknowns are even possible. That is a highly implausible modal claim, and one for which there is no demonstrable evidence. ◇Unknown does not need to prove that complete knowledge is false; it only needs to prove that it's logically possible for there to be unknowns, and that possibility never goes away. With ◇Unknown, it doesn't have to be necessary; it just has to be possible. Then, ◇Unknown's default logical stance is that complete knowledge cannot be proven, and its denial is always possible. Continually, if someone says they know everything, then this claim shows:

Formal knowledge of everything claim: $K_x(\forall p: T(p) \rightarrow K_x(p))$

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Except, if even that knowledge claim is unknowable in some possible world, then ◇Unknown applies to the claim of completeness itself. This creates a reflexive self-falsibility loop, which ◇Unknown can handle, but completeness cannot.

More falsifiers with theological and metaphysical claims. Most theologies, especially monotheistic ones, assert complete divine knowledge, perfect truth, and/or ultimate explanatory closure. For example, "God is omniscient", "All truth is revealed through scripture or revelation", and "There is a final answer to the question of being".

◇Unknown's Rebuttal: $\forall x \Diamond \exists p: T(p) \wedge \neg K_x(p)$

Even a divine or transcendent mind x may be such that there is a logically possible truth they do not (or cannot) know. Is this anti-theistic? Not necessarily. ◇Unknown does not deny the existence of a powerful or perfect being. It simply raises a logical boundary: that if we claim a being knows everything, we must explain how that being knows it knows, and that there's nothing left to know possibly. This leads to an infinite continuity, for example, "How do you know all that is knowable?"→"Because I know there is nothing else to know."→"But what if there's a mode of truth you are utterly oblivious to?" Even if there was a being that claimed to know all, in an infinite continuity, there are still possible modes of infinite continuity that are possibly totally outside of even divine knowledge; it's not like that being would know because it is outside of that being's understanding, and or knowledge.

Following the theological challenges come the metaphysical challenges. Questions arise, like, "*What is reality?*" "*What is the ground of being?*", and "*Are there necessary truths about existence?*". Metaphysical systems often attempt to map out all that exists, for example, dualism, materialism, and theism. ◇Unknown's role functions as a modal critique of closure. No matter how comprehensive a metaphysical system is, it's always logically possible that a truth exists outside of it, and or inside of it, that is unknown. Some examples could consist of ◇Unknown asking, "*Can you prove there's no possible truth beyond physicality?*", and a metaphysician could reply with, "*Reality is ultimately mental*", and ◇Unknown replies, "*It's possible that's only one aspect of a larger unknown structure.*" ◇Unknown calls these outlooks into a deeper expansion, one that aligns with objectively logical possibility rather than illogical absolute certainty.

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Two more outlooks that ◇Unknown challenges are Verificationism and Anti-Realism. Verificationism (Logical Positivism) holds that a statement is meaningful only if it can be empirically verified. For example, "*The meaning of a proposition is its verification method*". This denies metaphysical, theological, and or speculative statements as meaningless. Anti-Realism states there is no mind-independent reality; what is "real" is bound to what can be known, experienced, and or constructed within a framework, for example, perception, language, and science. An example that helps illustrate this outlook is that, in science, anti-realists could argue that we should not claim that atoms exist, but only that they're useful concepts in models. With ◇Unknown, Verificationism would state that since you can't verify it, it's meaningless. Except, ◇Unknown says that there may be truths that are real and logically possible, even if they can never be verified. The conflict is that Verificationism equates meaning with verification, which is very fragile, and ◇Unknown separates truth from knowability, which is a more logically "humble" approach. Verificationism fails as a logical universal principle under ◇Unknown, because it assumes that the space of truth is coextensive with the space of verifiability, which ◇Unknown directly denies. Secondly, for Anti-Realism, it says that since there is no epistemic access, there is no reality. On the contrary, ◇Unknown says that no epistemic access just means that it is an unknown reality. Anti-Realism's mistake is that it collapses truth into knowability, whereas ◇Unknown logically insists that truth can exceed knowability, even if unverifiable forever.

A final set of concerns that could arise, which will be put to rest, is the concern that ◇Unknown is self-contradictory. It was expressed within the proofs of ◇Unknown why this is not the case, but clarification will help form a stronger foundation. It is very important to consider that ◇Unknown does not assert any specific unknown proposition. This insight creates a difference between content and structure. ◇Unknown is a structural claim about epistemic limitation, not a claim about a particular fact. ◇Unknown is not asserting what facts must remain unknown, but asserts that there will always be unknowns. With our identification of structure over content, a similar objection to resolve is that ◇Unknown would need total knowledge of the possible space of unknowns. ◇Unknown is not making a survey claim but is stating a logical limitation. It's not that ◇Unknown surveys all spaces of unknowns, but rather highlights that

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non-closure cannot be ruled out, and that universal denial cannot logically be justified. If ◇Unknown is not understood properly, a paradox of the knower contradiction could arise. A contradiction in the sense that, since ◇Unknown asserts uncertainty and unpredictability within the unknown, hence ◇Unknown itself is not known nor certain, creating a self-reference paradox. Instead, ◇Unknown does not leave us in a hopeless fog of unknowns, but states that there possibly exists some proposition that is not known. This prevents reflexive indexing, semantic circularity, and a certain kind of liar structure. It is crucial to understand that ◇Unknown does not purport to identify any unknown truths, but it denies the validity of complete modal closure, as a structural modal claim about epistemic limitation.

4. A Deeper Look into Possibility

With certain fragile cases, it is important not to pair illogicality with impossibility too strictly. With contradictions and illogical claims, they are illogical, concluding implausibility. Except, even with contradictions and illogical statements, ◇Unknown still leaves room for these illogicalities to be, in a sort of way, "relevant". Even if a claim is found illogical, it is still logically possible to be valid/sound according to ◇Unknown. Except, do not rule out its illogicality. Even though ◇Unknown requires infinite epistemic openness, illogicalities and contradictions are still properly intact. Illogicalities and contradictions may still be logically possible, but because of their illogical makeup, the possibility that they are logical outside our knowledge is, most of the time, not worth deep consideration. Although not inherently impossible, being contradictory and illogical still holds, even with ◇Unknown. It is just that the more fragile these distinctions are, the more ◇Unknown challenges them.

On the subject of impossibility, does ◇Unknown make the impossible impossible? No, the relationship between ◇Unknown and impossibility, if not fully understood, can be seen by most as paradoxical. ◇Unknown says that impossibility could exist, yet also states that it couldn't. Well, which one should we choose? According to ◇Unknown, the existence of impossibility is the evidentially probable outcome. ◇Unknown declares that it is possible that the impossible doesn't exist. Still, it is more probable that the impossible exists, since certain axioms, like "◇*Unknown is not infinite*", are impossible because they contradict the logical law of ◇Unknown. Impossibility declares absolutely no possibility, so doesn't this contradict

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◇Unknown? Not at all. First of all, remember that this probability wouldn't state, "*Impossibility is absolutely true*". It would state, "*The most probable conclusion of the existence of impossibility is that it does exist evidentially from human knowledge*". Just because we are dealing with impossibility, it does not dismiss ◇Unknown's open-endedness.

On the other hand, with ◇Unknown, it does not erase impossibility, but instead says that there may be truths about what is impossible that we do not yet know. In other words, some things we conclude are impossible may actually be possible, and some things we conclude possible may actually be impossible, and we cannot know with absolute finality which is which. Additionally, ◇Unknown does not erase what is true. There are still definite, static truths of existence that are not affected by possibility in the perspective of objectivity, except that, with epistemic limitations, there will forever be the possibility of what we do not know about this objectivity. To put ◇Unknown and truth-values into perspective, from an objective viewpoint, ◇Unknown is a limitation of absolute, true knowledge on epistemic minds, and from an epistemic viewpoint, ◇Unknown is a limitation of our knowledge of objectivity. Impossibility most likely exists; we just cannot have final closure with the truth of it.

Entailment: In addition, let's say:

$I(p)$: "p is logically impossible"

$K_x(I(p))$: "agent x knows that p is impossible"

Then, ◇Unknown allows for: $\Diamond\exists p: I(p) \wedge \neg K_x(I(p))$

"It is possible that a proposition is impossible, and yet x does not know it is."

Also: $\Diamond\exists p: \neg I(p) \wedge \neg K_x(\neg I(p))$

"It is possible that a proposition is not impossible, and x does not know that either."

Therefore, ◇Unknown applies not just to truths, but to modal truths, including truths about what is impossible. ◇Unknown shows that our judgments of impossibility are themselves fallible. We may treat something as logically impossible, but that could be an epistemic error. For example, at one point, traveling faster than light was thought to be logically impossible, but later, under different physical models (such as the Alcubierre drive), it became a theoretical possibility, though still practically unreachable. ◇Unknown preserves the boundaries of impossibility, but insists that we may not have full access to where that boundary lies.

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What is the relationship between ◇Unknown and the limits of knowledge? The relationship between ◇Unknown and the limits of knowledge is not peripheral, but is definitional. While many philosophies of knowledge assume or infer that knowledge is incomplete, ◇Unknown formalizes this incompleteness as a logical law. It states that some truths remain unknown. This does not imply that knowledge is futile, but rather that it is permanently bound, not by error, but by structure.

Formal Relationship: $\forall x \Diamond \exists p \notin K_x \wedge T(p)$

"For any epistemic agent x, it is logically possible that there exists a truth that x does not know."

This does not say anything goes. It does not say that all things that are possible are to exist and or happen, but only that there are more possible truths than are currently known, including possibly truths about what is actually impossible. In doing so, ◇Unknown transforms the "limits of knowledge" from a vague epistemic constraint into a modal truth: the boundary between known and unknown is not fixed, and cannot be fully illuminated from within. Thus, the limits of knowledge are not signs of human failure or finitude; they are the terrain upon which all inquiry occurs.

A caution that needs to be taken with ◇Unknown is avoiding Skepticism and Fallibilism. Although ◇Unknown may share some elements with Skepticism and Fallibilism, they should not be categorized with ◇Unknown. Skepticism implies that we may not be able to know anything with certainty. Some or all knowledge claims are doubtful or unjustified. Although this is a possibility and partially ties into ◇Unknown's epistemic openness, it greatly overlooks probability and how we use evidence to differentiate unsupported possibilities from supported probabilities. This is why we have probability, to avoid pure skepticism. Fallibilism posits that we can know things, but all knowledge is potentially flawed, and that certainty is not necessary for justification. Fallibilism falls into a similar category to Skepticism, in the sense that probability with ◇Unknown undermines these outlooks. ◇Unknown does not cast doubt on whether knowledge is possible at all, but it affirms the existence and value of knowledge, and denies that it can ever be complete. In addition, ◇Unknown does not focus on the risk of error, but on the structural inevitability of incompleteness. It is not merely that our facts might be wrong, but that there are always true facts we do not yet and may never possess. Thus,

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◇Unknown is neither skeptical nor simply fallibilist: it is a modal law of epistemic openness grounded in logic rather than doubt and "humility" rather than despair.

4.1 In Accordance Across Truth-Preserving Fields

At this point, there is a logical boundary between distinguishing temporary unknowns from permanent unknowns. A temporary unknown is something currently beyond our grasp, but potentially within the reach of discovery; it is unknowable for now. ◇Unknown, however, mainly focuses on truths that are logically possible but may remain beyond the reach of any knower in any possible world. These are not merely hard to find; they may be structurally inaccessible, like truths outside the axioms of any complete formal system. Importantly, we cannot reliably distinguish the two from within our epistemic framework. ◇Unknown does not deny that temporary unknowns exist; it asserts that the space of unknowns is infinite, and that some of them are permanently unknowable not by proof, but by the logical openness of truth itself. With our epistemic frame, the line where permanent unknowns are distinguished from temporary unknowns cannot be formally found. ◇Unknown's infinite nature makes it so that we can't predict what we will know, and we can't predict what we cannot know. Unknowns that can be known are not always guaranteed to be known. Discovery is influenced by many variables in humans. So, what is discovered, and what is not, depends on what avenues a restricted epistemic mind chooses to pursue. ◇Unknown applies to all minds with knowledge, so we do not know whether a different mind knows an unknown that we do not know. This boundary is infinitely rooted within the laws of ◇Unknown, and trying to distinguish the two universally is implausible. This boundary constantly shifts and rearranges within the unknowns and knowns within certain minds. And even if the abstract hypothetical of knowing what others know and do not know were entertained, it would still be limited minds examining limited minds. Going as far as fabricating something of a '*hive mind*' would also show to be limited if in it were a composition of many minds, since whatever mind has any sort of limit is susceptible to ◇Unknown.

In relation, how does ◇Unknown work with mathematics and science? Well, it's structurally embedded within both fields. Mathematics might feel like the one area where everything should eventually be knowable, but that's not the case. According to Gödel's

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incompleteness theorems, in any consistent formal system powerful enough to describe arithmetic, there exist true statements that are unprovable within that system. ◇Unknown directly overlaps here.

Overlap Entailment: $\Diamond\exists p: T(p) \wedge \neg\vdash p$

"It is possible that there exists a true proposition that cannot be proven, and that may never be known."

In addition, there are mathematical questions for which no algorithm can give an answer at all, for example, the Halting Problem. Also, proven to be independent of Zermelo-Fraenkel set theory + choice, we can neither disprove it, a perfect example of ◇Unknown in action. So, ◇Unknown is already structurally embedded into mathematics via formal incompleteness, independence, and undecidability.

Subsequently, science is fundamentally about approximating truth through observation, modeling, and testing. Except, ◇Unknown shows that the field of possible truths is not bounded, not even by the laws we currently consider universal. Some key examples, as of now, are dark matter/energy, which comprise 95% of the known universe and are something we do not yet understand. Another example is the origin of the universe, and we may never be able to observe or explain the exact "cause" of the universe. We also lack a coherent framework for objectively measuring subjective experience, particularly the conscience. The measurement problem in quantum mechanics is a deep, unresolved question that may be epistemically unsolvable. Anthropic principle and or fine-tuning suggest that we may not be able to access all possible physical realities, only the one we happen to inhabit.

n: natural (deriving from scientific laws)

◇Unknown Scientific Form: $\Diamond\exists p \in P_n: T(p) \wedge \neg K_x(p)$

There are possibly true laws, constants, and or entities in nature that no agent can ever access. ◇Unknown does not mean that nothing can be known; it means that the total space of possible truths is larger than any epistemic system can contain, even in principle, so that knowns can exist, but every unknown cannot be known. That includes all true mathematical statements not derivable in any single system, all natural phenomena not observable due to scale, time, or logical limits, and all meta-level truths about truth itself. Thus, ◇Unknown is infinite in both mathematics and science.

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With the venture into science and focusing on the elements of unknowns, a principle that scientifically grounds ◇Unknown to a quantum level is the Heisenberg Uncertainty Principle. For context, the Heisenberg Uncertainty Principle states that it is fundamentally impossible to measure simultaneously and precisely both the position and momentum (or velocity) of a particle. For reference, the formula is provided:

Heisenberg Uncertainty Principle:

$\Delta x \Delta p \geq \frac{\hbar}{2}$ where $\hbar = \frac{h}{4\pi}$ (roughly)

The Heisenberg Uncertainty Principle establishes that uncertainty is a fundamental property of a limitation of physical properties. The uncertainty comes from the nature of a particle. Since particles act as waves, a particle does not have an exact location, but rather exists as a "*wave packet*", or a range of positions. This wave-particle duality shows that particles do not have well-defined trajectories, not in limited measurements, but in objectively defining the particle's nature. And when looking at measuring particles, the more precisely a particle's position is known, the less precisely its momentum can be known, and vice versa. Hence, when connecting it to ◇Unknown, what does this show? Using Heisenberg's Uncertainty Principle, let's say that we limit the case where position uncertainty approaches zero (*approaching zero uncertainty in* x). This would cause the momentum of the particle to exist with an infinite range of possible momenta (*wavelengths*), and vice versa.

Formal Representation: $\lim_{\Delta x \to 0} \Delta p \to \infty \vDash \neg\Diamond(\Delta x = 0 \wedge \Delta p = 0)$

It is impossible for both properties to be fully known simultaneously.

p_1**:** particle has precise position

p_2**:** particle has precise momentum

$\forall x \neg(K_x(p_1) \wedge (K_x(p_2))$

This does not arise from ignorance, but from state structure.

So, what we establish with Heisenberg's uncertainty principle is a substantial reinforcement of ◇Unknown's infinite nature. That, with the uncertain relationship between wave-particle duality, we observe that unknowns are necessary in quantum mechanics; hence, quantum mechanics provides an empirical instance in which structured reality itself enforces irreducible unknowns.

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Concerning a different topic, modal realism and metaphysical possibility are greatly connected with ◇Unknown. According to David Lewis, Modal Realism posits that all possible worlds are equally real, not just abstract or imagined. Possibility=something is true in some world, and Necessity=something is true in all worlds.

Formally: $\Diamond p$= there exists a world w such that p is true in w

$\Box p$= in all possible worlds, p is true

◇Unknown aligns with Modal Realism's Ontology. ◇Unknown claims that for every agent, there may be a truth they can never know. Similarly, Modal Realism says that possibilities exist concretely, even if we never access them. Thus, ◇Unknown could be interpreted as saying, "*There may exist possible worlds containing truths that are not accessible from this world, but those truths are still real.*" ◇Unknown does not need Lewisian Modal Realism to be valid, but it does gain ontological support from it. Modal Realism helps give ◇Unknown a concrete domain for its possibilities, and ◇Unknown adds to modal realism by highlighting the epistemic limits that persist even if the worlds are real. Following, metaphysical possibility refers to what is logically coherent and could exist in this world, and is broader than physical possibility. For example, it's physically implausible to travel faster than light, but metaphysically possible in some hypothetical reality. ◇Unknown uses metaphysical possibility as an optional avenue. Metaphysics holds that it is metaphysically possible that a truth exists that no one can know, and that this truth is real even if unknowable. ◇Unknown is a modal principle that does not need metaphysical possibility, but can sometimes be framed in terms of metaphysical possibility. ◇Unknown says that possibility is broader than knowledge, and, with this, some metaphysically consistent truths could be treated as never appearing in epistemic space.

4.2 Past Possibilities

As stated, ◇Unknown is great at working with possibilities for what is and what could be, but what about the past? Well, ◇Unknown says that there are facts about the past that are true, logically coherent, but permanently unknowable, perhaps due to data loss, non-survival, or absence of traces. For example, how did an unnamed human think or feel 85,000 years ago? There's almost certainly an answer to that; it was a true event in the world, but it may be forever unknowable, no records, no witnesses, and no mental access.

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ρ**:** originating from the past

◇**Unknown Past:** $\Diamond\exists p \in P_{\rho}(T(p) \wedge \neg K_x(p))$

"A true past possibly exists that is not and never will be known."

In deterministic models, the past and future are fixed in theory and fully accessible through physical laws. Except, ◇Unknown says, even in a deterministic universe, epistemic access to the past can be permanently lost. So, while Determinism assumes reversability, ◇Unknown maintains: "*Knowing the past≠the past being fixed.*" This undermines the idea that, in principle, everything can be known if the universe is Deterministic. But do not confuse this with a confirmation of free will. While ◇Unknown allows the possibility of free will, the reason past truths may never be known is generally due to "*we will never completely know whether or not this is what exactly happened.*" Another central philosophical assumption is that if a truth happened in the past, and the universe is closed, then it's knowable in principle. ◇Unknown completely disagrees and states that even if a past truth exists and is real, there's no guarantee any agent can ever access it. In relation to the past, information can be forgotten and epistemologically lost. By forgetting and losing this information, it becomes unknown again. Therefore, not only is ◇Unknown an infinite continuity concerning future possibilities, but also for past possibilities.

4.3 ◇Unknown for Application

How is ◇Unknown used? ◇Unknown is a vital concept in every field and is not bound to a specific instance. When ◇Unknown is a conclusion, the "*special*" characteristic of ◇Unknown is that it is its own conclusion. ◇Unknown is not dependent on empirical discovery; it is not deduced from lower-level propositions, and it acts more like a law, an axiom, or a foundational modal truth. ◇Unknown is a conclusion in the sense that it summarizes a necessary condition of epistemology. While ◇Unknown is a foundational conclusion, you can use it as a premise in logical or modal derivations, just like axioms are used in geometry and Peano arithmetic. ◇Unknown behaves like, "*Given this modal law, what else logically follows?*" That makes it both a conclusion about epistemic reality and a tool for limiting, shaping, or extending what else can be known or assumed. Another fundamental law of ◇Unknown is that when drawing connections between ◇Unknown and other atoms of PL languages, it can get quite trivial.

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Logical: $\forall x \Diamond \exists p \notin P \wedge K_x(T(p)) \vDash \Diamond(T(q) \wedge \neg K_x(q))$	◇Unknown by itself entails possibility
Illogical: $\forall x \Diamond \exists p \notin P \wedge K_x(T(p)) \vDash T(q)$	◇Unknown can never entail truth by itself
Logical: $\forall x \Diamond \exists p \notin P \wedge K_x(T(p)) \vDash P_{(\neg K_x(p))} > 0$	◇Unknown follows mathematical--truths to find probability
Illogical: $\forall x \Diamond \exists p \notin P \wedge K_x(T(p)) \vDash P_{(T(q))} = 1$	◇Unknown jumps from abstract unknowns--to empirical uncertainty by itself

The examples above show how ◇Unknown fundamentally works when applicable. ◇Unknown is best understood as a conclusion about the nature of knowability, in the same way that uncertainty is unavoidable, as a conclusion in quantum physics; some truths are unprovable, as a conclusion in Gödel's logic; and that possibility does not imply actuality, a modal distinction. ◇Unknown is both a modal conclusion and a logical constraint; it ends one inquiry (A*re unknowables possible? Yes*), and begins others (*Given ◇Unknown, what else logically follows?*). ◇Unknown entails modal openness, NOT modal resolution, and it defines an infinite logical horizon, NOT the map of what's definitively on it, which is why it can be quite trivial at times when employing ◇Unknown.

Are we at a point where we can permit ◇Unknown within our PL language *wffs*?

5. Identification of Assumptions

When looking into the meta of ◇Unknown, it is necessary to distinguish the meta from the non-meta of ◇Unknown because they differ in their applications. This differentiating of the two is heavily induced by ◇Unknown challenging itself in its own application. The main question this difference is seeking to answer is, "*What assumptions are made when defining ◇Unknown, and can we avoid them?*" Based on the assumptions that ◇Unknown sometimes creates, two different versions of ◇Unknown are presented: Relative ◇Unknown and Universal ◇Unknown.

Universal ◇Unknown: Universal ◇Unknown is the meta application of ◇Unknown. When taking ◇Unknown in its most vast and wide spanned form, we come to the conclusion that "*complete objective knowledge is impossible for any epistemic mind to achieve due to the infinite*

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nature of limited receptive knowledge." With this, we make assumptions that make ◇Unknown quite fragile. We are assuming that every epistemic mind is unable to have complete objective knowledge, and that total epistemic closure in any case is illogical and unreasonable. This claim cannot be logical in a full sense, since ◇Unknown goes against this conclusion. We are trying to make a universal assumption in our limited knowledge, and ◇Unknown sobers this conclusion by countering it with a statement along the lines of, "*You cannot make a universal conclusion of complete knowledge without having it yourself.*" This meta-argument cannot be accepted due to the limits of our own knowledge and our ignorance of every epistemic mind that will ever be.

Universal ◇Unknown Statement:

S**:** any epistemic system

$\forall S \exists p \Diamond (T(p) \wedge \neg K_S(p))$

Relative ◇Unknown: Relative ◇Unknown is the non-meta application of ◇Unknown. After reviewing Universal ◇Unknown and identifying its fragile assumptions, Relative ◇Unknown is particularly important to consider, as it rejects these assumptions and provides a much more logically grounded and polished version of ◇Unknown. As stated before, the assumptions we make with Universal ◇Unknown are that every epistemic mind is unable to have complete objective knowledge and that epistemic closure is illogical and unreasonable. With Relative ◇Unknown, we rather only apply ◇Unknown to epistemic minds we already identify as limited, and do not assume that our limited observations span into the unknown. We prevent fragile assumptions and instead accept that, since we are limited in knowledge, the possibilities of alien/divine logic could not be limited, and that complete objective knowledge could be possible in principle, we just may never be able to achieve knowledge of this, since we are limited epistemic minds. Relative ◇Unknown is a much stronger application of ◇Unknown and helps guide a much more cautious approach to the unknown.

Relative ◇Unknown Statement:

L_x: epistemically limited agent

$\forall x (L_x \rightarrow \exists p \Diamond (T(p) \wedge \neg K_S(p)))$

5.1 Qualification of Law and Theory

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Taking both of our versions of ◊Unknown into account, Relative ◊Unknown is on a foundation where it can be qualified as a fundamental law of logic pertaining to epistemology. Universal ◊Unknown is a theory, where we should not treat it as a law, applicable universally, but rather a logical insight for alien/divine minds and systems, and what qualifications we must recall in the logical epistemic analysis of minds and systems.

◊Unknown as a Law: ◊Unknown as a law of logic is what was defined as Relative ◊Unknown. The reason Relative ◊Unknown is a law is that it stays logically grounded and does not extend into theoretical fields, like Universal ◊Unknown. Relative ◊Unknown is domain-restricted; it does not overstep its boundaries. We only apply it when a mind or system is epistemically limited, which is every system/mind to our knowledge, so at least in our case, looking at an objectively open standpoint, Relative ◊Unknown is always applicable in practice as of now. Not only that, but Relative ◊Unknown is necessary when dealing with unknowns. Without Relative ◊Unknown, the navigation and distinction of barriers, limits, and unknowns in general would be disoriented, and unknowns could be majorly mistreated and misused. Even across variation in epistemic systems and minds, Relative ◊Unknown does not change. For any limited mind/system, it is always the case that they will never have complete objective knowledge. That is applicable universally to limited minds/systems, not just to particular ones.

◊Unknown as a Theory: ◊Unknown as a theory of logic is what was defined as Universal ◊Unknown. The reason Universal ◊Unknown is a theory is due to its nature of surpassing our own barriers of knowledge and making assumptions that are not logically grounded. Universal ◊Unknown goes against itself in a sense where, since we're assuming that this applies universally beyond our knowledge, ◊Unknown brings up an equally valid assumption that it possibly doesn't apply universally beyond our knowledge. With Universal ◊Unknown, in a sense, we are overstepping our boundaries. How can we state that ◊Unknown applies to all minds if we do not know every mind, and whether or not they have limits? Clearly, we cannot. Although while granting leniency for alien/divine minds, it accepts that there may be unknown minds that could provide evidence somewhat in contrast with ◊Unknown, this is not evidence of such minds.

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While ◇Unknown and many other logically grounded arguments previously reviewed provide evidence to build fundamental frameworks for ◇Unknown, taking the identification of assumptions, an important barrier to logic that supports our identification of assumptions, and supports ◇Unknown's challenges of limited knowledge, needs to be demonstrated.

5.2 Hume's Law: The Particular/Universal Barrier

The particular/universal barrier, closely related to Hume's Law, states that a set of only particular premises cannot entail a universal conclusion. In comparison of ◇Unknown and this barrier, we are not talking about the classic *is-ought* formulation alone. With the particular/universal barrier and ◇Unknown, this is not a derivation, but a showing that the same inferential barrier applies. With this showing, some caution must be taken, since ◇Unknown is not an ethical principle, hence why we are not concerned with the *is-ought* of Hume's Law. The alignment of the particular/universal barrier and ◇Unknown is as follows:

Particular/Universal Barrier	**◇Unknown**
Finite observations	Finite knowledge states
Universal generalization	Modal exhaustiveness
Missing entailment	Possibility residue

By this means, "*the leap from 'all observed' to 'all possible' is unjustified.*" ◇Unknown simply generalizes this insight beyond induction. Especially when looking into arguments against ◇Unknown, they require a universal claim over possibly space, yet according to the Humean particular/universal barrier, no finite set of particulars can entail such a universal (*our particulars being epistemic limitations*). Hence, the denial of ◇Unknown is unjustified.

Minimal Formalization:

Here we have agent x's knowledge: $K_x=\{p_1,p_2,...p_n\}$

The denial of ◇Unknown requires: $K_x \vdash \forall p(P(K_x(p)))$

But by Humean barrier: $\{p_1,...p_n\} \nvdash \forall p$

Hence: $\neg\vdash\neg\Diamond(\exists p(T(p)\wedge\neg K_x(p)))$

Which is equivalent to: $\vdash\Diamond(\exists p(T(p)\wedge\neg K_x(p)))$

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A very important distinction to make is why this does not just collapse into Hume's Law. This is because Hume's Law concerns the illegitimacy of certain inferential transitions, while ◇Unknown concerns the modal residue that remains once such transitions are blocked. ◇Unknown can be understood as the modal analog of Hume's particular/universal barrier: a necessary residue that persists whenever finite knowledge is used to deny universal possibility.

6. Conclusion

In conclusion, ◇Unknown is not a problem, nor a dilemma, but is a foundation of modal possibility that challenges us further to confirm and demonstrate truth within any conclusion. ◇Unknown reminds us that the boundaries of what can be known are not static, but are structurally open. As both an epistemic law and a modal principle, ◇Unknown formalizes the logical possibility that truth exceeds cognition, that for any knower, there may always be truths that remain forever out of reach. This is not an appeal to ignorance, but is a disciplined recognition of the infinite incompleteness of epistemic access. Whether applied to science, mathematics, metaphysics, theology, or history, ◇Unknown acts as a principle of objective logic and a tool for intellectual "humility" and ontological realism. It preserves space for truths that exist even when unverifiable, untraceable, and or conceptually inaccessible. In doing so, it resists premature closure, invites deeper inquiry, and affirms the logical "dignity" of the unknown. It does not undermine truth; it cautions its mystery.

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